Spectrograms, Vowels & Formants

Dr. Josef Fruehwald

```
source(here::here("_defaults.R"))
```

```
Loading required package: sysfonts
```

```
Loading required package: showtextdb
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.2.1     ✓ readr     2.2.0
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ lubridate 1.9.5     ✓ tibble    3.3.1
✓ purrr     1.2.2     ✓ tidyr     1.3.2
```

```
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(tidynorm)
library(ggdensity)
library(ggdiagram)
library(geomtextpath)
library(tuneR)
```

```

Attaching package: 'tuneR'

The following object is masked from 'package:ggdiagram':
```

```
    bind
```

```
library(seewave)
```

```

Attaching package: 'seewave'

The following object is masked from 'package:ggdiagram':

    rescale

The following object is masked from 'package:lubridate':

    duration

The following object is masked from 'package:readr':

    spec
```

```
library(patchwork)
library(scales)
```

```

Attaching package: 'scales'

The following object is masked from 'package:seewave':

    rescale

The following object is masked from 'package:ggdiagram':

    rescale

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```
library(praatpicture)
```

```

Attaching package: 'praatpicture'

The following object is masked from 'package:colorspace':

    specplot
```

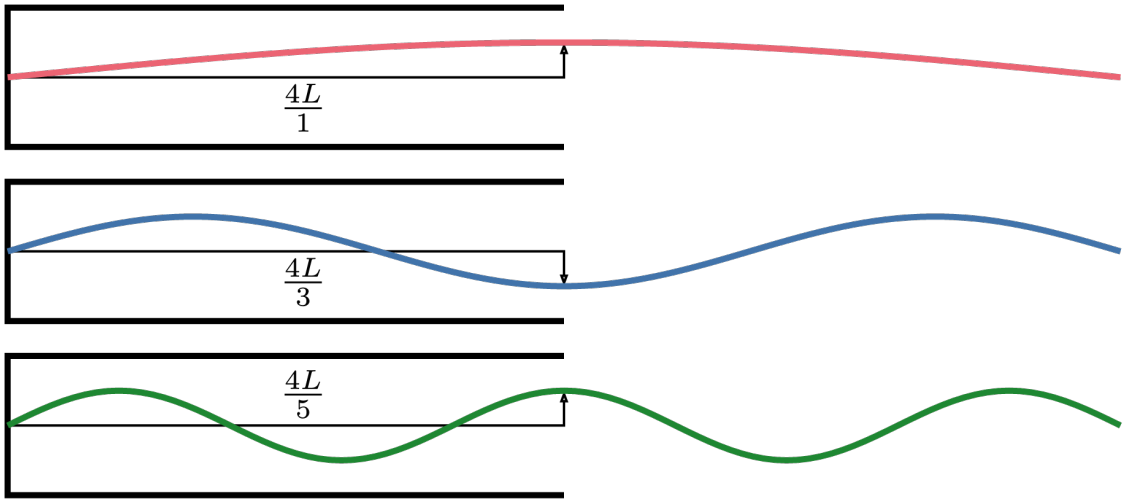
```
library(tinytable)
library(mgcv)
```

```
Loading required package: nlme

Attaching package: 'nlme'

The following object is masked from 'package:dplyr':

    collapse

This is mgcv 1.9-4. For overview type '?mgcv'.
```

```
wav2spect <- function(path, window = 0.005, step = 0.002, dyn = -50) {
  wav <- readWave(path)
  win_len <- window * wav@samp.rate
  hop_len <- step * wav@samp.rate
  overlap <- ((win_len - hop_len) / win_len) * 100

  spect <- wav |>
    spectro(
      # window length, in terms of samples
      wl = win_len,
      # window overlap
      ovlp = overlap,
      # don't plot the result
      plot = F
    )

  # set the colnames and rownames
  colnames(spect$amp) <- spect$time
  rownames(spect$amp) <- spect$freq
```

```
  spect$amp |>
    # coerce the row names to a column
    as_tibble(rownames = "freq") |>
    # pivot to long format
    pivot_longer(
      # all columns except freq
      -freq,
      names_to = "time",
      values_to = "amp"
    ) |>
    # since they were names before,
    # freq and time need conversion to numeric
    mutate(
      freq = as.numeric(freq),
      time = as.numeric(time)
    ) |>
    mutate(
      amp_floor = case_when(
        amp < dyn ~ dyn,
        TRUE ~ amp
      )
    )
}
```

Formants (resonant frequencies)



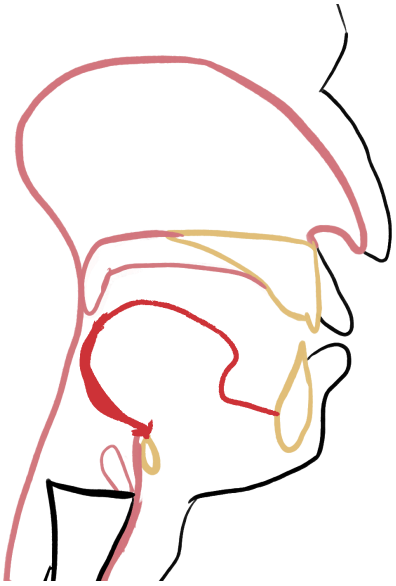
Formants (resonant frequencies)

If L = 15cm, then

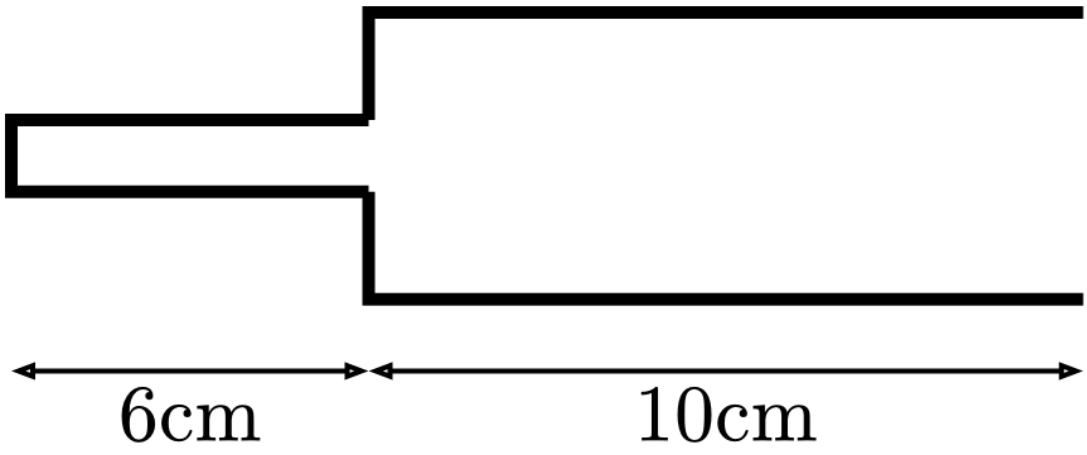
Resonance	Wavelength	Frequency
1	$\frac{4L}{1} = 60\ \text{cm}$	$\frac{34300\frac{cm}{s}}{60\ \text{cm}} = 571\ \text{Hz}$
2	$\frac{4L}{3} = 20\ \text{cm}$	$\frac{34300\frac{cm}{s}}{20\ \text{cm}} = 1715\ \text{Hz}$
3	$\frac{4L}{5} = 12\ \text{cm}$	$\frac{34300\frac{cm}{s}}{12\ \text{cm}} = 2858\ \text{Hz}$

Vowels - More Complex

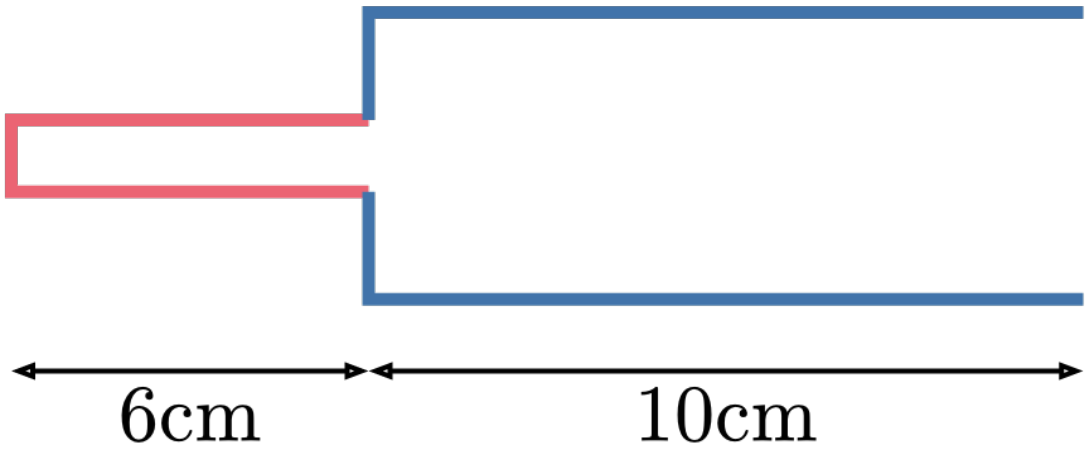
A vowel like [ɑ] is no longer a simple tube.



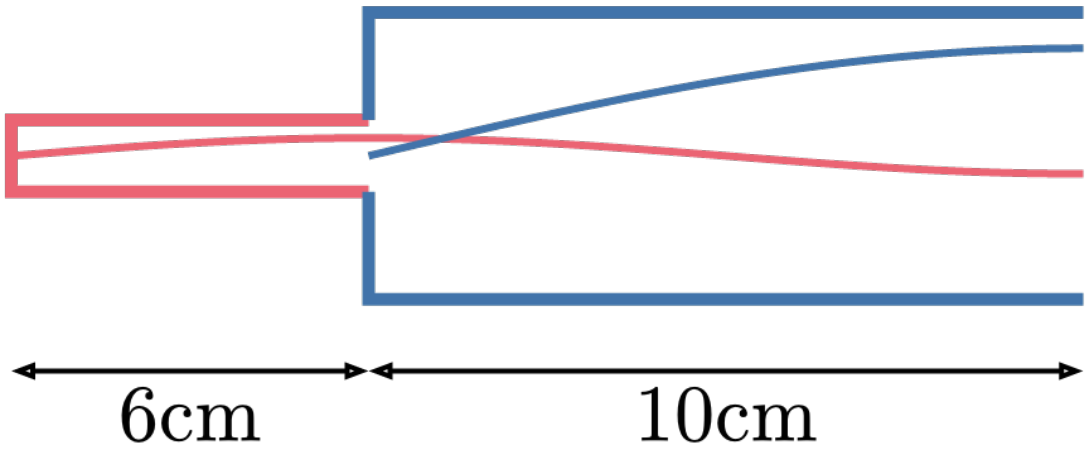
Vowels



Vowels



Vowels



Vowels

```
tibble(
  Resonance = 1:2,
  Frequency = (((2*Resonance) -1) * 34300)/ (4 * 6)
) |>
  tt() |>
  format_tt(
    digits = 1
  ) |>
  style_tt(
    j = 1:2,
    align = "r"
  )
tibble(
  Resonance = 1:2,
  Frequency = (((2*Resonance) -1) * 34300)/ (4 * 10)
) |>
  tt() |>
  format_tt(
    digits = 1
  ) |>
  style_tt(
    j = 1:2,
    align = "r"
  )
```

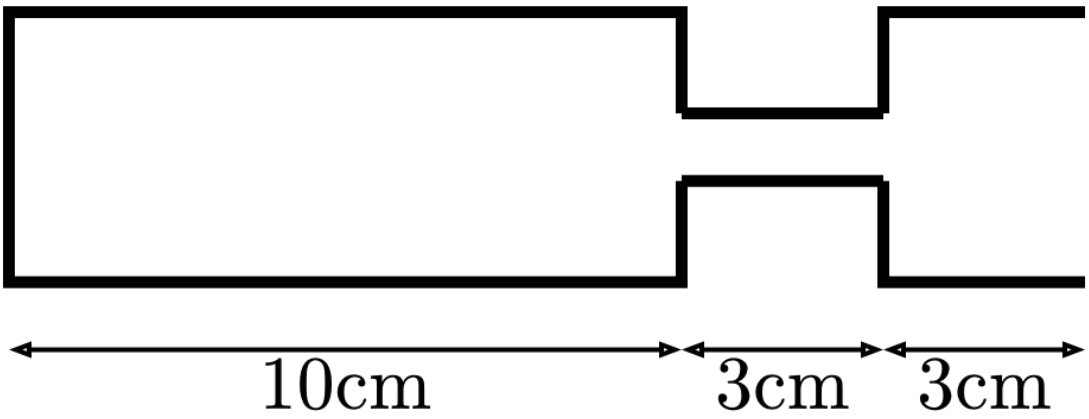
Resonance	Frequency
1	1429
2	4288

Resonance	Frequency
1	858
2	2572

Vowels

A vowel like [iː] is even more complex

Vowels

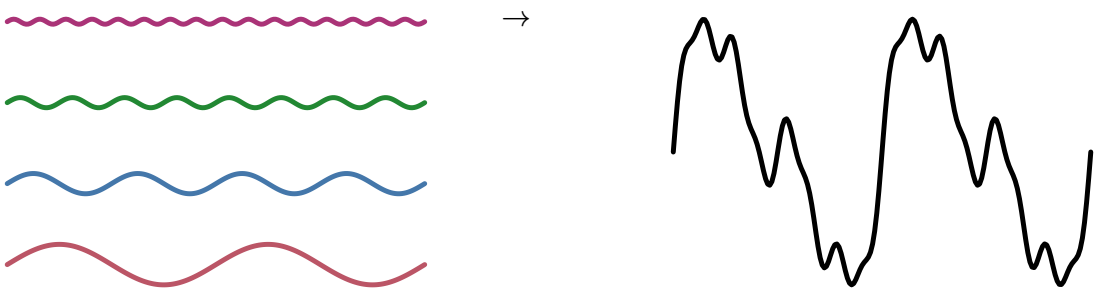


Creating Complex Sounds

```
make_sinex <- function(x=0, freq = 1, t = 0){
  sin((freq/2) * x) * cos(t * freq) * (1/freq)
}
```

```
expand_grid(
  y = 0,
  x = seq(0, 2*pi, length = 200),
  t = 0,
  freq = c(4, 8, 16, 32)
) |>
  mutate(
    .by = freq,
    y = make_sinex(x, freq, t),
    freq_f = factor(freq)
  ) ->
  many_wav

many_wav |>
  ggplot(
    aes(x, y + as.numeric(freq_f))
  ) +
  geom_line(
    aes(
      color = factor(freq)
    ),
    linewidth = 2,
```

```
    lineend = "round"
  ) +
  guides(color = "none") +
  theme_void()
many_wav |>
  summarise(
    .by = x,
    y = sum(y)
  ) |>
  ggplot(
    aes(x, y)
  ) +
  geom_line(
    linewidth = 2,
    lineend = "round"
  ) +
  theme_void()
```

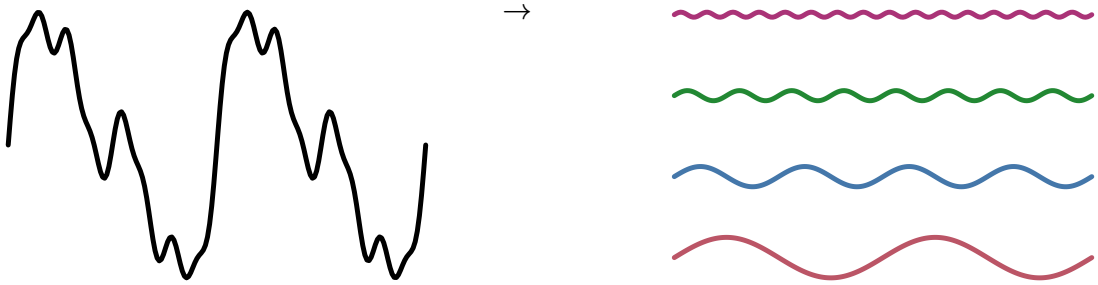


Decomposing complex sounds

a.k.a. Fourier Analysis

```
many_wav |>
  summarise(
    .by = x,
    y = sum(y)
  ) ->
  complex_wav

complex_wav |>
  ggplot(
    aes(x, y)
  ) +
  geom_line(
```

```
    linewidth = 2,
    lineend = "round"
  ) +
  theme_void()
many_wav |>
  ggplot(
    aes(x, y + as.numeric(freq_f))
  ) +
  geom_line(
    aes(
      color = factor(freq)
    ),
    linewidth = 2,
    lineend = "round"
  ) +
  guides(color = "none") +
  theme_void()
```



Decomposing Complex Signals

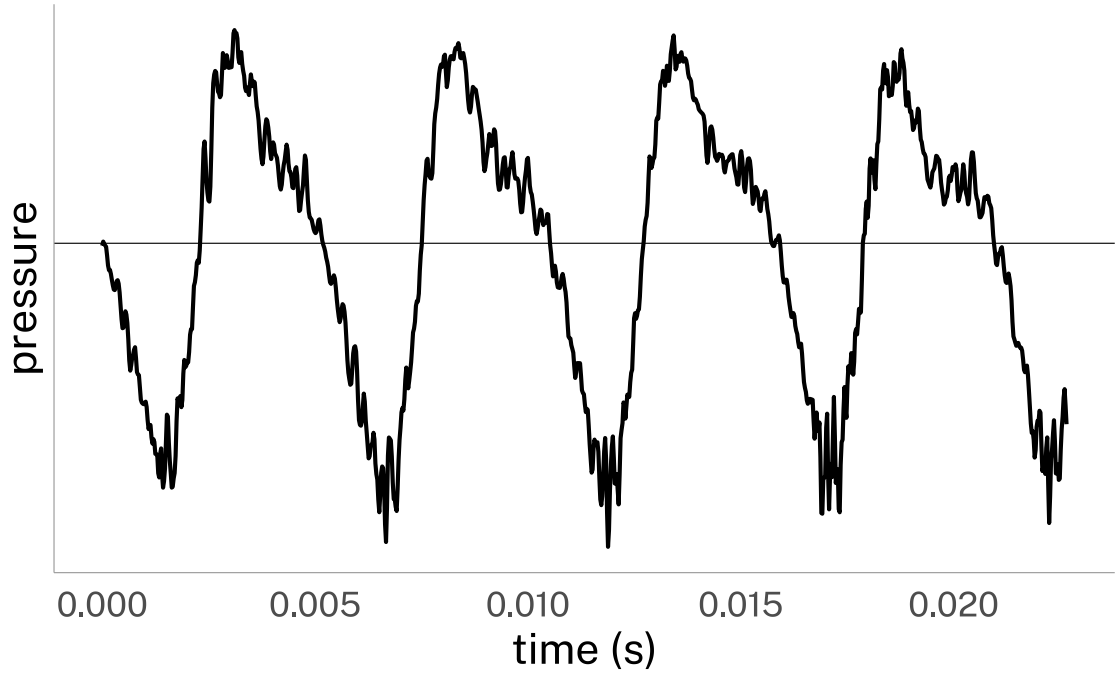
An actual wavform of [i]

```
iy <- readWave(here::here("data/audio/iy.wav"))
```

```
tibble(
  wav = iy@left[2001:3000]
) |>
  mutate(
    time = row_number()-1
  ) ->
  iy_wav1
```

```
tibble(
  pressure = iy@left,
  samp_rate = iy@samp.rate
) |>
  mutate(
    samp = row_number() - 1,
    time = samp/samp_rate,
    window = floor(samp/1000)
  ) |>
  filter(
    .by = window,
    n() == 1000
  ) |>
  mutate(
    .by = window,
    wsamp = row_number() - 1,
    mid_time = median(time),
    mag = abs(fft(pressure)),
    freq = wsamp/(1000/samp_rate)
  ) ->
  ffts
```

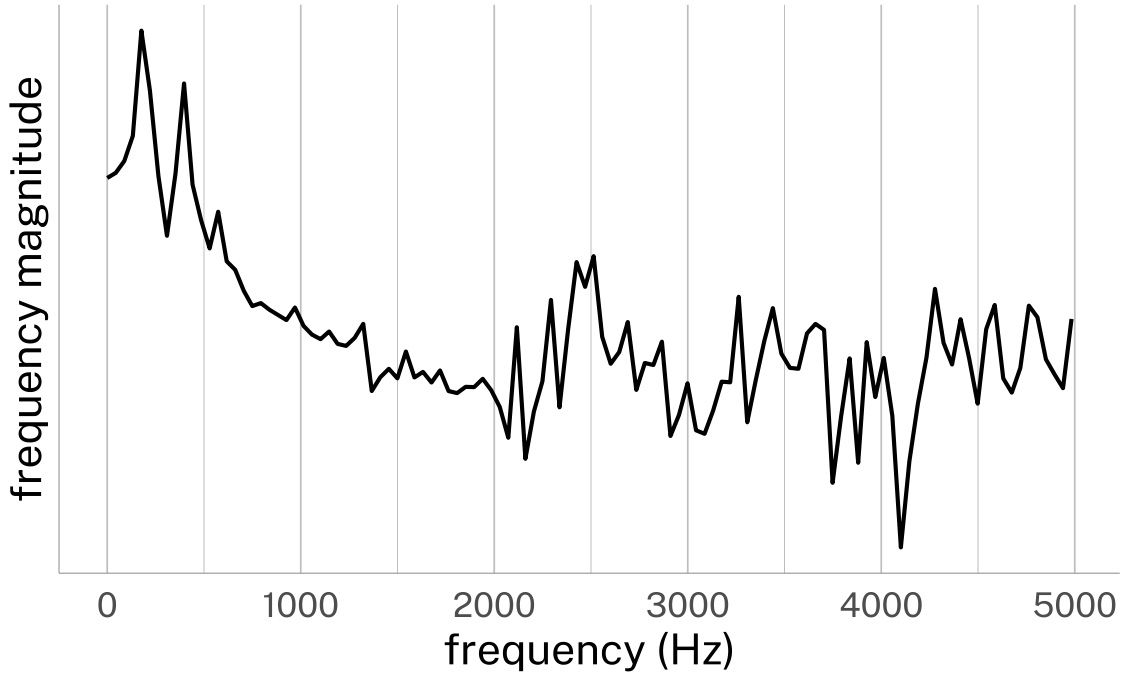
```
ffts |>
  filter(
    window == 3
  ) |>
  ggplot(
    aes(wsamp/samp_rate, pressure)
  ) +
  geom_hline(
    yintercept = 0,
    linewidth = 0.2
  ) +
  geom_line() +
  labs(
    x = "time (s)"
  ) +
  theme_sub_axis_y(
    text = element_blank()
  )
```



Decomposing Complex Signals

```
ffts |>
  filter(
    window == 3,
    freq <= 5000
  )  |>
  ggplot(
    aes(freq, mag)
  ) +
  geom_line() +
  scale_y_log10() +
  labs(
    x = "frequency (Hz)",
    y = "frequency magnitude"
  ) +
  theme_sub_panel(
    grid.major.x = element_line(
      linewidth = 0.3,
      color = "grey"
    ),
    grid.minor.x = element_line(
      linewidth = 0.1,
      color = "grey"
```

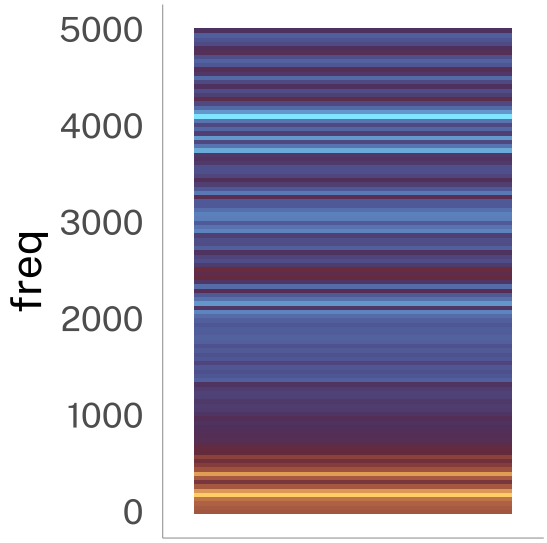
```
    )
  ) +
  theme_sub_axis_y(
    text = element_blank()
  )
```



Decomposing Complex Signals

```
ffts |>
  filter(
    window == 3,
    freq <= 5000
  ) |>
  ggplot(
    aes(
      x = "0",
      y = freq
    )
  ) +
  geom_tile(
    aes(fill = log(mag))
  ) +
  scale_fill_continuous(
```

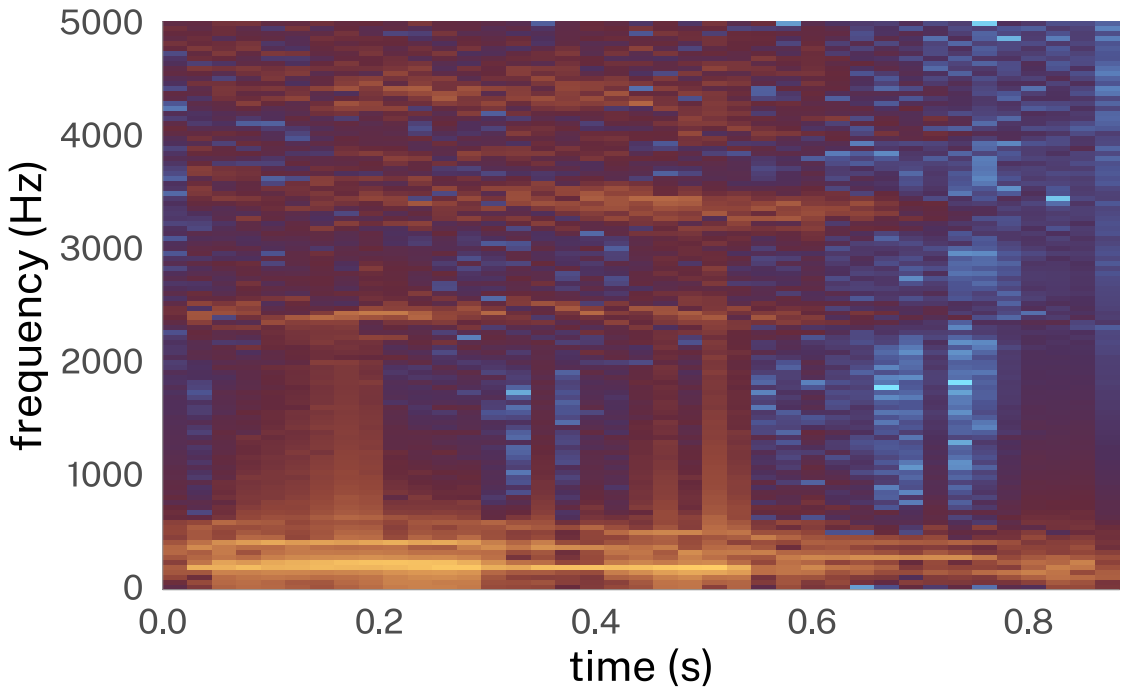
```
    trans = "reverse"
  ) +
  guides(
    fill = "none"
  ) +
  theme_no_x()
```



A Spectrogram

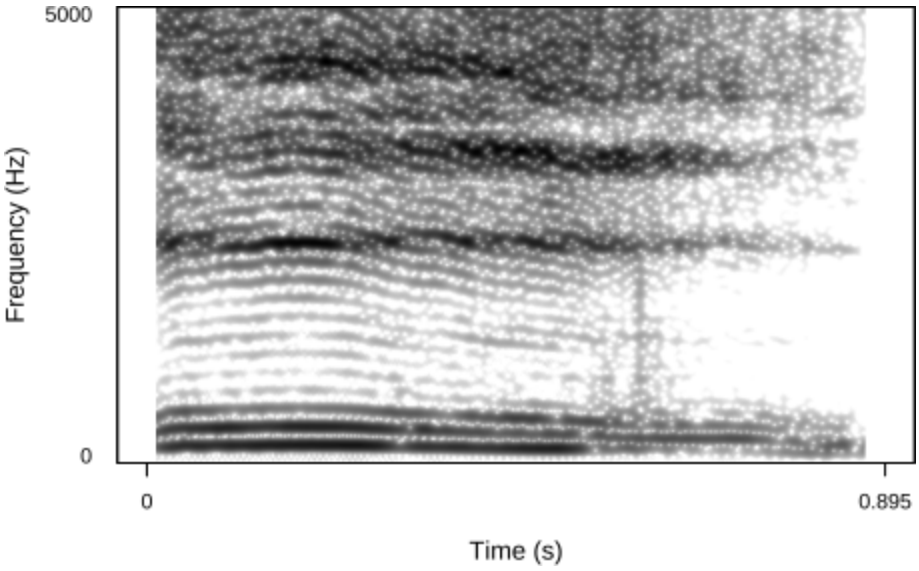
```
ffts |>
  filter(
    freq <= 5000
  ) |>
  ggplot(
    aes(
      x = mid_time,
      y = freq
    )
  ) +
  geom_tile(
    aes(fill = log(mag))
  ) +
  scale_fill_continuous(
    trans = "reverse"
  ) +
  scale_y_continuous(
    expand = expansion(0)
```

```
  ) +
  scale_x_continuous(
    expand = expansion(0)
  ) +
  guides(
    fill = "none"
  ) +
  labs(
    x = "time (s)",
    y = "frequency (Hz)"
  )
```



A Spectrogram

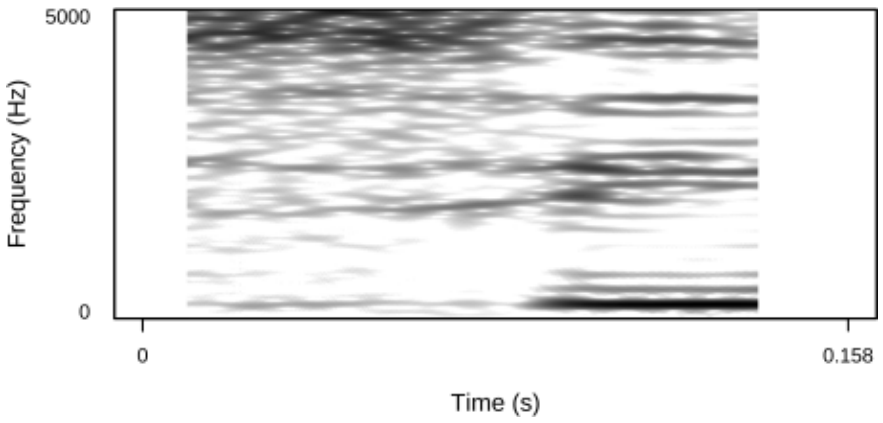
```
praatpicture(
  here::here("data/audio/iy.wav"),
  frames = c("spectrogram"),
  spec_windowLength = 1000/iy@samp.rate
)
```



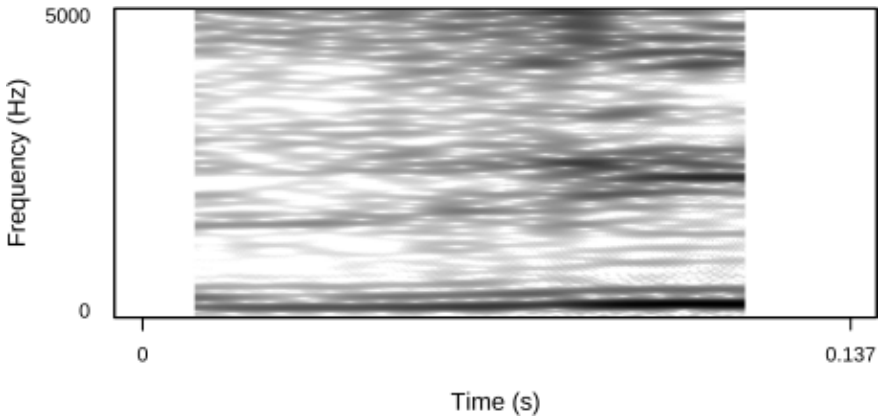
Spectrograms

```
praatpicture(
  here::here("data/audio/si.wav"),
  frames = c("spectrogram"),
  spec_windowLength = 0.02
)
praatpicture(
  here::here("data/audio/zi.wav"),
  frames = c("spectrogram"),
  spec_windowLength = 0.02
)
```

[si]



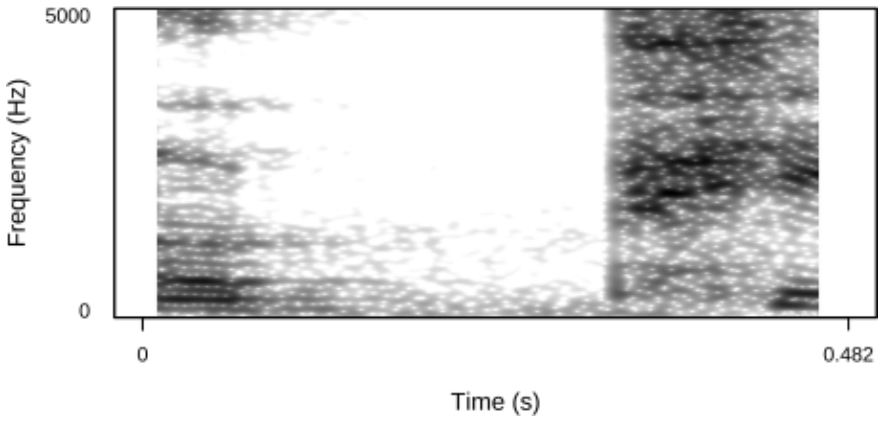
[zi]



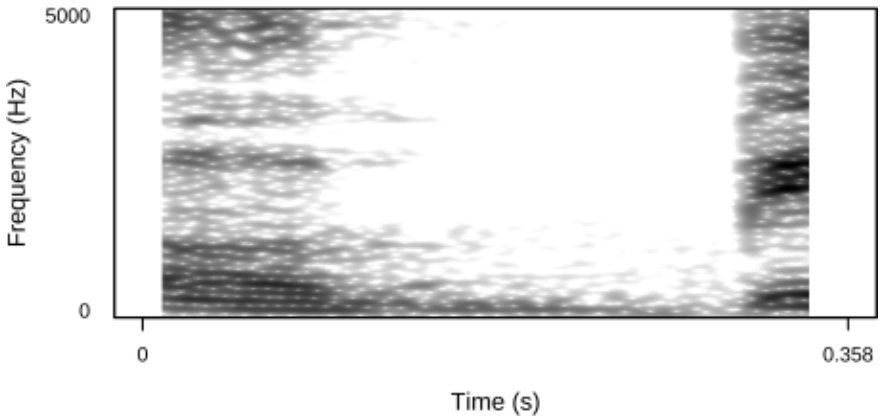
Spectrograms

```
praatpicture(
  here::here("data/audio/api.wav"),
  frames = c("spectrogram"),
  spec_windowLength = 0.02
)
praatpicture(
  here::here("data/audio/abi.wav"),
  frames = c("spectrogram"),
  spec_windowLength = 0.02
)
```

[əpʰi]



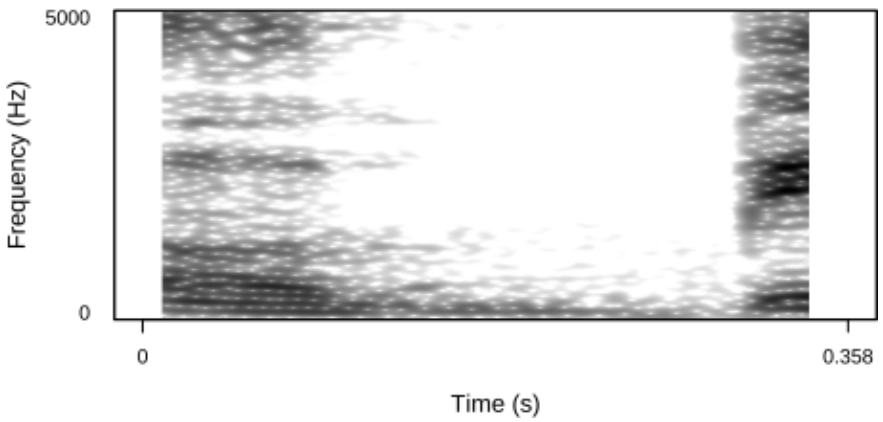
[əbi]



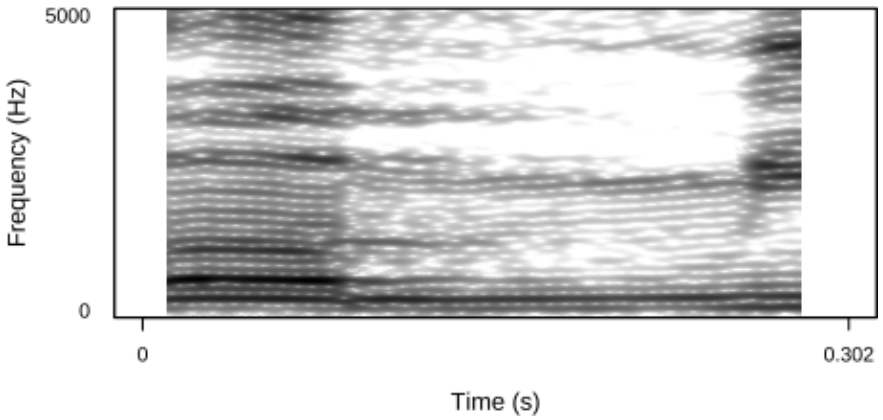
Spectrograms

```
praatpicture(
  here::here("data/audio/abi.wav"),
  frames = c("spectrogram"),
  spec_windowLength = 0.02
)
praatpicture(
  here::here("data/audio/ami.wav"),
  frames = c("spectrogram"),
  spec_windowLength = 0.02
)
```

[əbi]



[əmi]



Spectrograms

```
wav2spect(here::here("data/audio/aa.wav"), dyn = -35)->
  aa_spect

wav2spect(here::here("data/audio/ae.wav"), dyn = -35)->
  ae_spect

wav2spect(here::here("data/audio/eh.wav"), dyn = -35)->
  eh_spect

wav2spect(here::here("data/audio/e.wav"), dyn = -35)->
  e_spect

wav2spect(here::here("data/audio/iy.wav"), dyn = -35)->
  iy_spect

tibble(
  param = 0:5,
  F1 = c(633.1/sqrt(2), 24.9, 16, 23.7, 34.1, 32),
  F2 = c(988/sqrt(2), 35.7, 73.5, 40.9, 82.1, 40.1),
  F3 = c(2675.4/sqrt(2), 35.1, 40, 39, -9.1, 13.7)
) |>
  slice(0:3) |>
  reframe_with_idct(
    F1:F3
  ) |>
  mutate(
    .time = .time-1,
    .time = (.time/max(.time))*max(aa_spect$time)
  ) |>
```

```
  pivot_longer(
    F1:F3
  ) ->
  aa_track
```

```
Warning in reframe_with_idct(slice(tibble(param = 0:5, F1 = c(633.1/sqrt(2), :
There is was no grouping provided.
ℹ You may want to provide `.by` with a speaker id column.
```

```

```

```
ℹ No `.param_col` provided.
ℹ Assuming `.data` is arranged by parameter.
```

```
Warning: Using `by = character()` to perform a cross join was deprecated in
dplyr 1.1.0.
ℹ Please use `cross_join()` instead.
ℹ The deprecated feature was likely used in the tidynorm package.
  Please report the issue at <https://github.com/JoFrhwld/tidynorm/issues>.
```

```
tibble(
  param = 0:2,
  F1 = c(828/sqrt(2), -35.3, -10.7),
  F2 = c(1547.2/sqrt(2), 29.1, 24.7),
  F3 = c(2156.2/sqrt(2), -165.6, 168.9)
) |>
  reframe_with_idct(
    F1:F3
  ) |>
  mutate(
    .time = .time-1,
    .time = (.time/max(.time))*max(ae_spect$time)
  ) |>
  pivot_longer(
    F1:F3
  ) ->
  ae_track
```

```
Warning in reframe_with_idct(tibble(param = 0:2, F1 = c(828/sqrt(2), -35.3, :
There is was no grouping provided.
ℹ You may want to provide `.by` with a speaker id column.
```

```
ℹ No `.param_col` provided.
ℹ Assuming `.data` is arranged by parameter.
```

```
tibble(
  F1 = c(586.3/sqrt(2), 102.5, -2.7),
  F2 = c(1882.1/sqrt(2), 4.6, -12),
  F3 = c(2593.9/sqrt(2), 20.7, 11.9)
) |>
  reframe_with_idct(
    F1:F3
  ) |>
  mutate(
    .time = .time-1,
    .time = (.time/max(.time))*max(eh_spect$time)
  ) |>
  pivot_longer(
    F1:F3
  ) ->
  eh_track
```

```
Warning in reframe_with_idct(tibble(F1 = c(586.3/sqrt(2), 102.5, -2.7), : There
is was no grouping provided.
ℹ You may want to provide `.by` with a speaker id column.
```

```
ℹ No `.param_col` provided.
ℹ Assuming `.data` is arranged by parameter.
```

```
tibble(
  F1 = c(421.9/sqrt(2), 34.5, 6.7),
  F2 = c(2152.7/sqrt(2), -78.4, 14.2),
  F3 = c(2524.5/sqrt(2), -2.6, 23.7)
) |>
  reframe_with_idct(
    F1:F3
  ) |>
  mutate(
    .time = .time-1,
    .time = (.time/max(.time))*max(e_spect$time)
  ) |>
  pivot_longer(
```

```
    F1:F3
  ) ->
  e_track
```

```
Warning in reframe_with_idct(tibble(F1 = c(421.9/sqrt(2), 34.5, 6.7), F2 =
c(2152.7/sqrt(2), : There is was no grouping provided.
ℹ You may want to provide `.by` with a speaker id column.
```

```

ℹ No `.param_col` provided.
ℹ Assuming `.data` is arranged by parameter.
```

```
tibble(
  F1 = c(288.6/sqrt(2), 11.5, 0),
  F2 = c(2410.4/sqrt(2), 7.6, -0.2),
  F3 = c(3461.8/sqrt(2), 33.9, 89.6)
) |>
  reframe_with_idct(
    F1:F3
  ) |>
  mutate(
    .time = .time-1,
    .time = (.time/max(.time))*max(iy_spect$time)
  ) |>
  pivot_longer(
    F1:F3
  ) ->
  iy_track
```

```
Warning in reframe_with_idct(tibble(F1 = c(288.6/sqrt(2), 11.5, 0), F2 =
c(2410.4/sqrt(2), : There is was no grouping provided.
ℹ You may want to provide `.by` with a speaker id column.
```

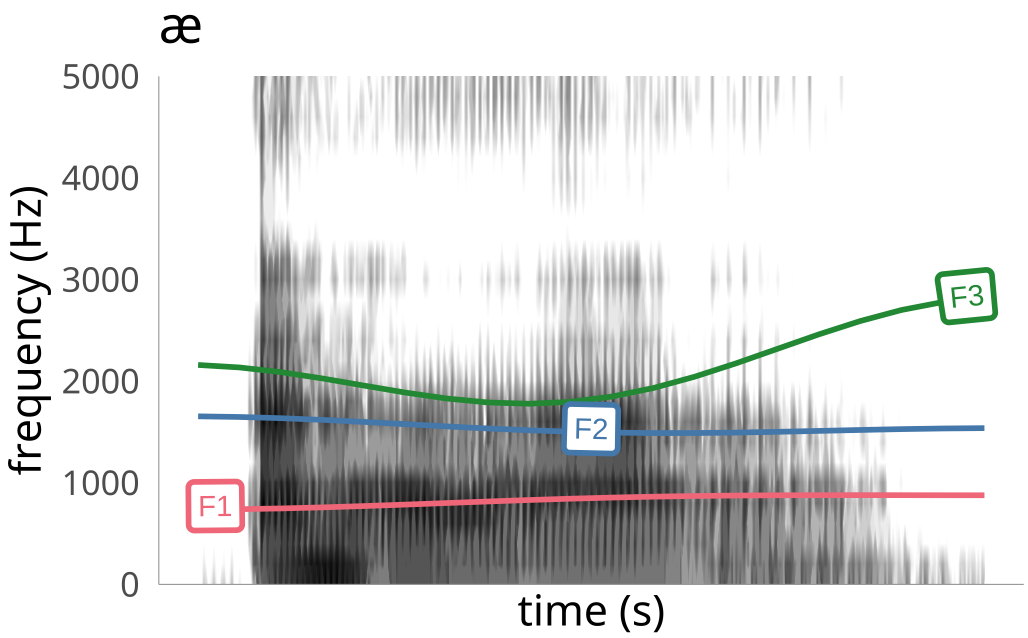
```

ℹ No `.param_col` provided.
ℹ Assuming `.data` is arranged by parameter.
```

```
aa_spect |>
  ggplot(aes(time, freq*1000))+
  stat_contour(
    aes(
```

```
      z = amp_floor,
      fill = after_stat(level)
    ),
    geom = "polygon",
    bins = 300
  ) +
  geom_labelline(
    data = aa_track,
    aes(
      x = .time,
      y = value,
      color= name,
      label = name,
      hjust = name
    ),
    linewidth = 1,
    show.legend = F
  ) +
  guides(fill = "none")+
  labs(
    x = "time (s)",
    y = "frequency (Hz)"
  ) +
  scale_fill_gradient(
    low = "white", high = "black"
  ) +
  scale_color_manual(
    values = c(ptol_red, ptol_blue, ptol_green)
  )+
  scale_y_continuous(
    expand = expansion(0)
  )+
  coord_cartesian(
    ylim = c(0,5000)
  )  +
  theme_sub_axis_x(
    text = element_blank()
  ) ->
  aa_plot
```

```
ae_spect |>
  ggplot(aes(time, freq*1000))+
  stat_contour(
    aes(
```

```
      z = amp_floor,
      fill = after_stat(level)
    ),
    geom = "polygon",
    bins = 300
  ) +
  geom_labelline(
    data = ae_track,
    aes(
      x = .time,
      y = value,
      color= name,
      label = name,
      hjust = name
    ),
    linewidth = 1,
    show.legend = F
  ) +
  guides(fill = "none")+
  labs(
    x = "time (s)",
    y = "frequency (Hz)",
    title = "æ"
  ) +
  scale_fill_gradient(
    low = "white", high = "black"
  ) +
  scale_color_manual(
    values = c(ptol_red, ptol_blue, ptol_green)
  )+
  scale_y_continuous(
    expand = expansion(0)
  )+
  coord_cartesian(
    ylim = c(0,5000)
  )  +
  theme_sub_axis_x(
    text = element_blank()
  ) +
  theme(
    text = element_text(family = "Noto Sans")
  ) ->
  ae_plot
ae_plot
```




```
eh_spect |>
  ggplot(aes(time, freq*1000))+
    stat_contour(
      aes(
        z = amp_floor,
        fill = after_stat(level)
      ),
      geom = "polygon",
      bins = 300
    ) +
    geom_labelline(
      data = eh_track,
      aes(
        x = .time,
        y = value,
        color= name,
        label = name,
        hjust = name
      ),
      linewidth = 1,
      show.legend = F
    ) +
    guides(fill = "none")+
    labs(
      x = "time (s)",
```

```
      y = "frequency (Hz)",
      title = "ε"
    ) +
    scale_fill_gradient(
      low = "white", high = "black"
    ) +
    scale_color_manual(
      values = c(ptol_red, ptol_blue, ptol_green)
    )+
    scale_y_continuous(
      expand = expansion(0)
    )+
    coord_cartesian(
      ylim = c(0,5000)
    )  +
    theme_sub_axis_x(
      text = element_blank()
    )+
    theme(text = element_text(family = "Noto Sans"))->
  eh_plot
```

```
e_spect |>
  ggplot(aes(time, freq*1000))+
    stat_contour(
      aes(
        z = amp_floor,
        fill = after_stat(level)
      ),
      geom = "polygon",
      bins = 300
    ) +
    geom_labelline(
      data = e_track,
      aes(
        x = .time,
        y = value,
        color= name,
        label = name,
        hjust = name
      ),
      linewidth = 1,
      show.legend = F
    ) +
    guides(fill = "none")+
```

```
    labs(
      x = "time (s)",
      y = "frequency (Hz)",
      title = "e"
    ) +
    scale_fill_gradient(
      low = "white", high = "black"
    ) +
    scale_color_manual(
      values = c(ptol_red, ptol_blue, ptol_green)
    )+
    scale_y_continuous(
      expand = expansion(0)
    )+
    coord_cartesian(
      ylim = c(0,5000)
    ) +
    theme_sub_axis_x(
      text = element_blank()
    ) +
    theme(
      text = element_text(family = "Noto Sans")
    ) ->
  e_plot
```

```
iy_spect |>
  ggplot(aes(time, freq*1000))+
    stat_contour(
      aes(
        z = amp_floor,
        fill = after_stat(level)
      ),
      geom = "polygon",
      bins = 300
    ) +
    geom_labelline(
      data = iy_track,
      aes(
        x = .time,
        y = value,
        color= name,
        label = name,
        hjust = name
      ),
```

```
      linewidth = 1,
      show.legend = F
    ) +
    guides(fill = "none")+
    labs(
      x = "time (s)",
      y = "frequency (Hz)",
      title = "i"
    ) +
    scale_fill_gradient(
      low = "white", high = "black"
    ) +
    scale_color_manual(
      values = c(ptol_red, ptol_blue, ptol_green)
    )+
    scale_y_continuous(
      expand = expansion(0)
    )+
    coord_cartesian(
      ylim = c(0,5000)
    ) +
    theme_sub_axis_x(
      text = element_blank()
    )  +
    theme(
      text = element_text(family = "Noto Sans")
    )->
  iy_plot
```

Vowels

```
ae_plot + eh_plot + e_plot + iy_plot +
  plot_layout(axes = "collect", nrow = 1)
```

Vowels

```
wav2spect(here::here("data/audio/aa.wav"), dyn = -35)->
  aa_spect

tibble(
  F1 = c(594.6/sqrt(2), 20.3, 26.7),
  F2 = c(829.6/sqrt(2), 33.9, 27),
  F3 = c(2632.1/sqrt(2), -152.8, -7.1)
) |>
  reframe_with_idct(
    F1:F3
  ) |>
  mutate(
    .time = .time-1,
    .time = (.time/max(.time))*max(aa_spect$time)
  ) |>
  pivot_longer(
    F1:F3
  ) ->
  aa_track
```

```
Warning in reframe_with_idct(tibble(F1 = c(594.6/sqrt(2), 20.3, 26.7), F2 =
c(829.6/sqrt(2), : There is was no grouping provided.
ℹ You may want to provide `.by` with a speaker id column.
```

```

```

```
ℹ No `.param_col` provided.
ℹ Assuming `.data` is arranged by parameter.
```

```
aa_spect |>
  ggplot(aes(time, freq*1000))+
    stat_contour(
      aes(
        z = amp_floor,
        fill = after_stat(level)
      ),
      geom = "polygon",
      bins = 300
    ) +
    geom_labelline(
      data = aa_track,
      aes(
        x = .time,
        y = value,
        color= name,
        label = name,
        hjust = name
      ),
      linewidth = 1,
      show.legend = F
    ) +
    guides(fill = "none")+
    labs(
      x = "time (s)",
      y = "frequency (Hz)",
      title = "a"
    ) +
    scale_fill_gradient(
      low = "white", high = "black"
    ) +
    scale_color_manual(
      values = c(ptol_red, ptol_blue, ptol_green)
    )+
    scale_y_continuous(
      expand = expansion(0)
    )+
    coord_cartesian(
      ylim = c(0,5000)
    ) +
    theme_sub_axis_x(
      text = element_blank()
    )  +
    theme(
```

```
      text = element_text(family = "Noto Sans")
    ) ->
  aa_plot
```

```
wav2spect(here::here("data/audio/ao.wav"), dyn = -35)->
  ao_spect

tibble(
 F1 = c(448.7/sqrt(2), 53, -16.2),
 F2 = c(621.5/sqrt(2), -32.7, -28.6),
 F3 = c(2705.4/sqrt(2), 22.7, -20.6)
) |>
  reframe_with_idct(
    F1:F3
  ) |>
  mutate(
    .time = .time-1,
    .time = (.time/max(.time))*max(ao_spect$time)
  ) |>
  pivot_longer(
    F1:F3
  ) ->
  ao_track
```

```
Warning in reframe_with_idct(tibble(F1 = c(448.7/sqrt(2), 53, -16.2), F2 =
c(621.5/sqrt(2), : There is was no grouping provided.
ℹ You may want to provide `.by` with a speaker id column.
```

```

```

```
ℹ No `.param_col` provided.
ℹ Assuming `.data` is arranged by parameter.
```

```
ao_spect |>
  ggplot(aes(time, freq*1000))+
    stat_contour(
      aes(
        z = amp_floor,
        fill = after_stat(level)
      ),
      geom = "polygon",
      bins = 300
```

```
    ) +
    geom_labelline(
      data = ao_track,
      aes(
        x = .time,
        y = value,
        color= name,
        label = name,
        hjust = name
      ),
      linewidth = 1,
      show.legend = F
    ) +
    guides(fill = "none")+
    labs(
      x = "time (s)",
      y = "frequency (Hz)",
      title = "ɔ"
    ) +
    scale_fill_gradient(
      low = "white", high = "black"
    ) +
    scale_color_manual(
      values = c(ptol_red, ptol_blue, ptol_green)
    )+
    scale_y_continuous(
      expand = expansion(0)
    )+
    coord_cartesian(
      ylim = c(0,5000)
    ) +
    theme_sub_axis_x(
      text = element_blank()
    )  +
    theme(
      text = element_text(family = "Noto Sans")
    ) ->
  ao_plot
```

```
wav2spect(here::here("data/audio/o.wav"), dyn = -35)->
  o_spect

tibble(
  F1 = c(367.5/sqrt(2), 58.3, -34.2),
```

```
  F2 = c(716.1/sqrt(2), 5.9, 48.2),
  F3 = c(2355.1/sqrt(2), 145.1, -36.3)
) |>
  reframe_with_idct(
    F1:F3
  ) |>
  mutate(
    .time = .time-1,
    .time = (.time/max(.time))*max(o_spect$time)
  ) |>
  pivot_longer(
    F1:F3
  ) ->
  o_track
```

```
Warning in reframe_with_idct(tibble(F1 = c(367.5/sqrt(2), 58.3, -34.2), : There
is was no grouping provided.
ℹ You may want to provide `.by` with a speaker id column.
```

```
ℹ No `.param_col` provided.
ℹ Assuming `.data` is arranged by parameter.
```

```
o_spect |>
  ggplot(aes(time, freq*1000))+
    stat_contour(
      aes(
        z = amp_floor,
        fill = after_stat(level)
      ),
      geom = "polygon",
      bins = 300
    ) +
    geom_labelline(
      data = o_track,
      aes(
        x = .time,
        y = value,
        color= name,
        label = name,
        hjust = name
```

```
      ),
      linewidth = 1,
      show.legend = F
    ) +
    guides(fill = "none")+
    labs(
      x = "time (s)",
      y = "frequency (Hz)",
      title = "o"
    ) +
    scale_fill_gradient(
      low = "white", high = "black"
    ) +
    scale_color_manual(
      values = c(ptol_red, ptol_blue, ptol_green)
    )+
    scale_y_continuous(
      expand = expansion(0)
    )+
    coord_cartesian(
      ylim = c(0,5000)
    ) +
    theme_sub_axis_x(
      text = element_blank()
    )  +
    theme(
      text = element_text(family = "Noto Sans")
    ) ->
  o_plot
```

```
wav2spect(here::here("data/audio/u.wav"), dyn = -35)->
  u_spect



tibble(
  F1 = c(270.7/sqrt(2), 37.7, -47.5),
  F2 = c(671.3/sqrt(2), -30.1, -10.4),
  F3 = c(2231.2/sqrt(2), 6.8, -10.3)
) |>
  reframe_with_idct(
    F1:F3
  ) |>
  mutate(
```

```
    .time = .time-1,
    .time = (.time/max(.time))*max(u_spect$time)
  ) |>
  pivot_longer(
    F1:F3
  ) ->
  u_track
```

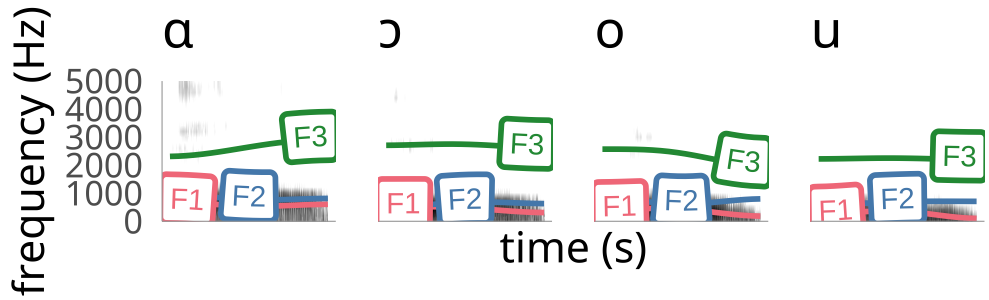
```
Warning in reframe_with_idct(tibble(F1 = c(270.7/sqrt(2), 37.7, -47.5), : There
is was no grouping provided.
ℹ You may want to provide `.by` with a speaker id column.
```

```
ℹ No `.param_col` provided.
ℹ Assuming `.data` is arranged by parameter.
```

```
u_spect |>
  ggplot(aes(time, freq*1000))+
    stat_contour(
      aes(
        z = amp_floor,
        fill = after_stat(level)
      ),
      geom = "polygon",
      bins = 300
    ) +
    geom_labelline(
      data = u_track,
      aes(
        x = .time,
        y = value,
        color= name,
        label = name,
        hjust = name
      ),
      linewidth = 1,
      show.legend = F
    ) +
    guides(fill = "none")+
    labs(
      x = "time (s)",
```

```
      y = "frequency (Hz)",
      title = "u"
    ) +
    scale_fill_gradient(
      low = "white", high = "black"
    ) +
    scale_color_manual(
      values = c(ptol_red, ptol_blue, ptol_green)
    )+
    scale_y_continuous(
      expand = expansion(0)
    )+
    coord_cartesian(
      ylim = c(0,5000)
    ) +
    theme_sub_axis_x(
      text = element_blank()
    )  +
    theme(
      text = element_text(family = "Noto Sans")
    ) ->
  u_plot
```

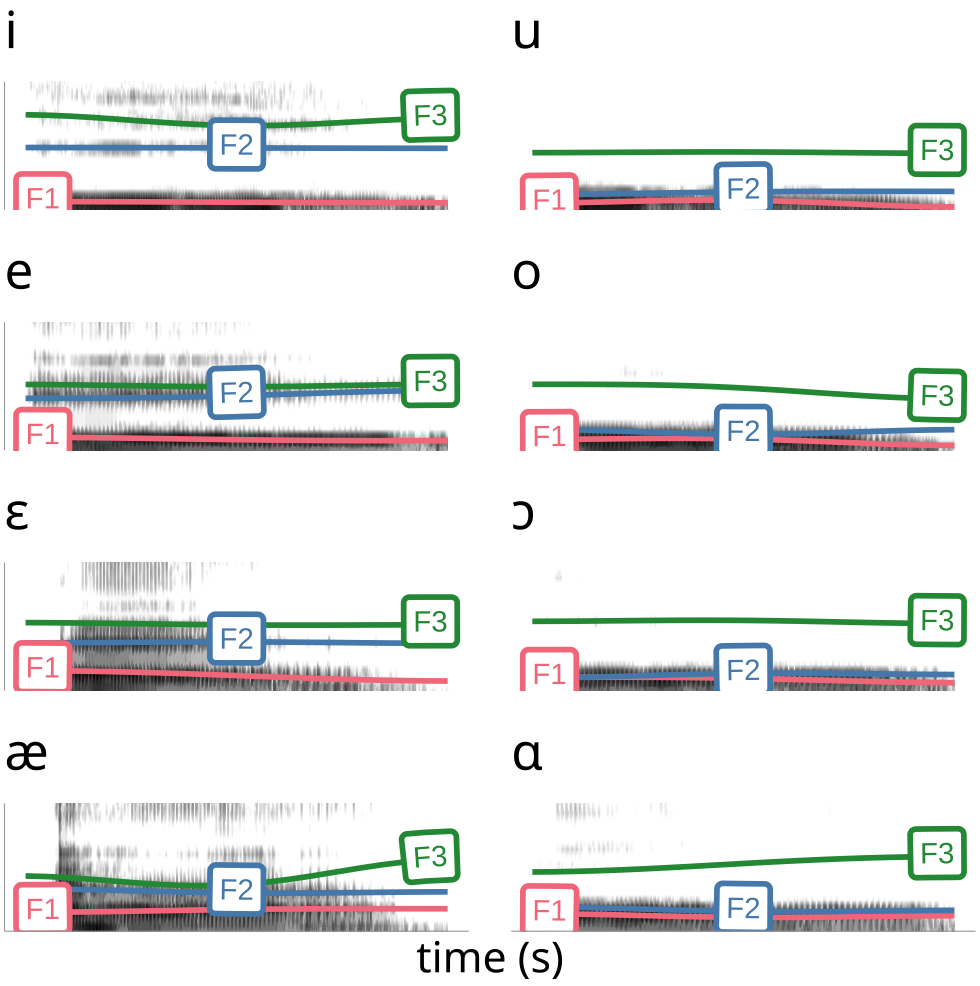
```
aa_plot + ao_plot + o_plot + u_plot +
  plot_layout(axes = "collect", nrow = 1)
```



Vowels

```
iy_plot + u_plot +
  e_plot + o_plot +
  eh_plot + ao_plot +
```

```
ae_plot + aa_plot +
  plot_layout(axes = "collect", ncol= 2) &
theme_no_y()
```

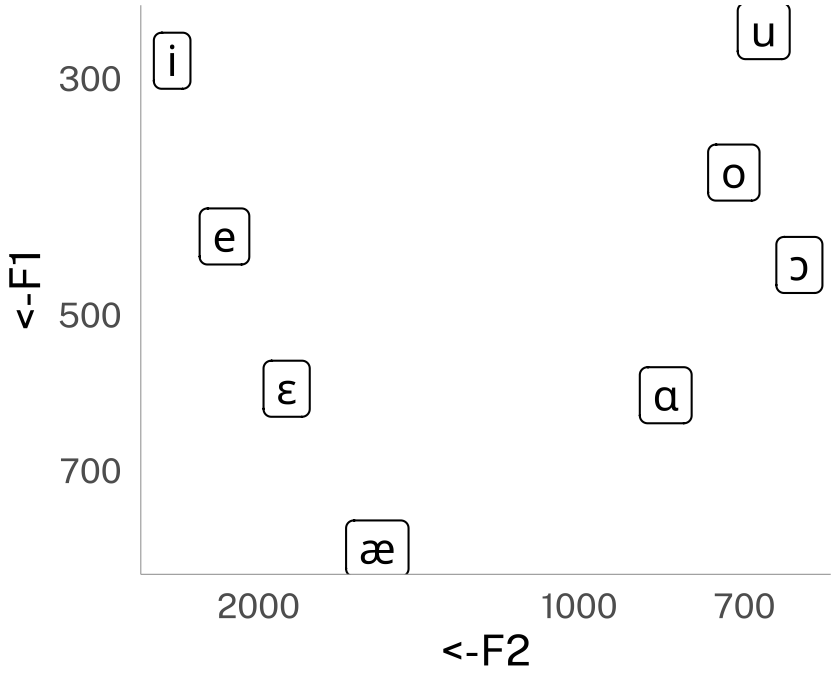


Vowel Space

```
list(
  aa_track |>
    mutate(vowel = "ɑ"),
  ae_track |>
    mutate(vowel = "æ"),
  eh_track |>
    mutate(vowel = "ɛ"),
```

```
  e_track |>
    mutate(vowel = "e")  ,
  iy_track |>
    mutate(vowel = "i"),
  ao_track |>
    mutate(vowel = "ɔ") ,
  o_track |>
    mutate(vowel = "o") ,
  u_track |>
    mutate(vowel = "u")
) |>
  list_rbind() |>
  pivot_wider(
    names_from = name,
    values_from = value
  )->
  all_track

all_track |>
  summarise(
    .by = vowel,
    across(F1:F2, mean)
  ) ->
  vowel_mean
```

```
vowel_mean |>
  ggplot(
    aes(F2, F1)
  )+
  geom_label(
    aes(label = vowel),
    family = "Noto Sans"
  )+
  labs(
    x = "<-F2",
    y = "<-F1"
  )+
  scale_x_continuous(transform = c("log10", "reverse"))+
  scale_y_continuous(transform = c("log10", "reverse"))+
  coord_fixed()
```



Vowel Space

```
anae <- read_rds("~/OneDrive - University of Kentucky/Corpora/ANAE/rds_data/anaeUNNORM.rds")
```

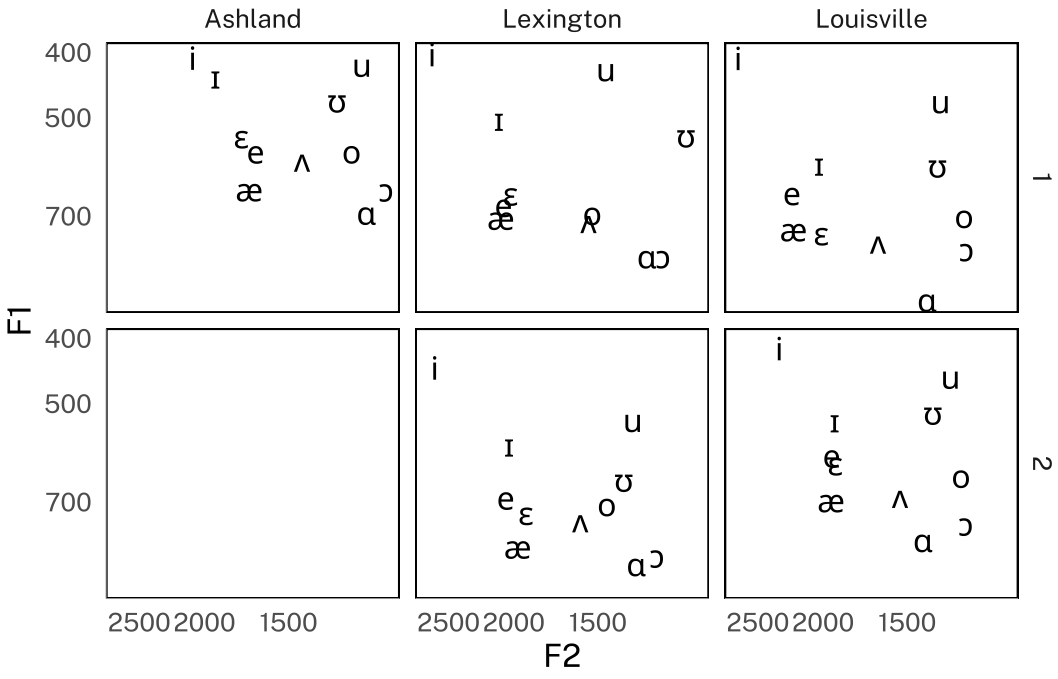
```
anae |>
  filter(
    State == "KY",
    !FolSeg %in% c("M", "N", "NG")
  ) |>
  filter(
    VClass %in% c(
      "iy","ey","i","e","ae",
      "o","uh","ow","uw","oh",
      "u"
    )
  ) |>
  mutate(
    vclass = case_match(
```

```
      VClass,
      "iy" ~ "i",
      "ey" ~ "e",
      "i" ~ "ɪ",
      "e" ~ "ɛ",
      "ae" ~ "æ",
      "o" ~ "ɑ",
      "uh" ~ "ʌ",
      "ow" ~ "o",
      "uw" ~ "u",
      "oh" ~ "ɔ",
      "u" ~ "ʊ"
    )
  ) |>
   mutate(
    .by = City2,
    speaker = consecutive_id(TS)
  ) ->
  ky
```

```
Warning: There was 1 warning in `mutate()`.
ℹ In argument: `vclass = case_match(...)`.
Caused by warning:
! `case_match()` was deprecated in dplyr 1.2.0.
ℹ Please use `recode_values()` instead.
```

```
ky |>
  summarise(
    .by = c(TS, City2, VClass, vclass),
    across(F1:F2, mean)
  ) |>
  mutate(
    .by = City2,
    speaker = as.numeric(as.factor(TS))
  ) |>
  mutate(
    .by = City2,
    speaker = consecutive_id(speaker)
  ) |>
  ggplot(
    aes(F2, F1)
  )+
  geom_text(
```

```
    aes(label = vclass),
    family = "Noto Sans"
  )+
  scale_y_continuous(
    transform = c("log10", "reverse")
  )+
  scale_x_continuous(
    transform = c("log10", "reverse")
  )+
  coord_fixed()+
  theme_sub_panel(
    border = element_rect()
  ) +
  facet_grid(speaker~City2)
```

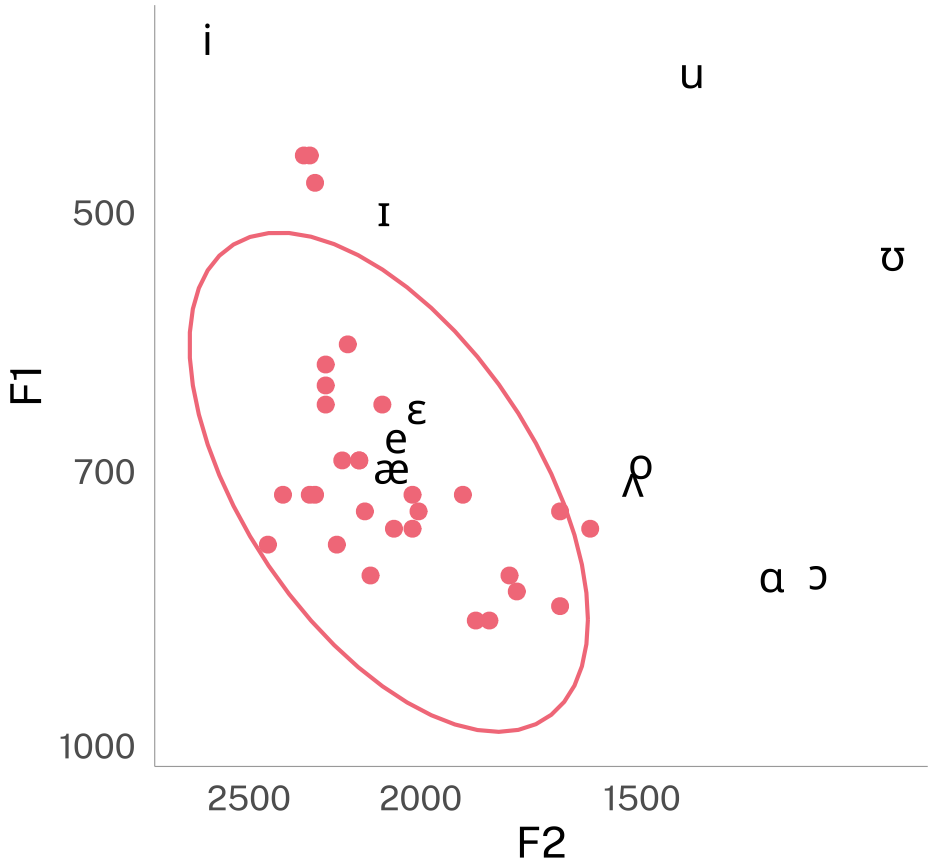


Vowel Space

```
make_mean <- function(df){
  df |>
    summarise(
      .by = matches("[^xy]"),
      across(
        c(x,y),
```

```
        mean
      )
    )
}
```

```
ky |>
  filter(
    City2 == "Lexington",
    speaker == 1
  ) -> lex1

lex1 |>
  ggplot(
    aes(F2, F1)
  )+
  geom_text(
    aes(label = vclass),
    stat = "manual",
    fun = make_mean,
    family = "Noto Sans"
  ) +
  geom_point(
    data = lex1 |> filter(VClass == "ae"),
    color = ptol_red
  )+
  stat_ellipse(
    data = lex1 |> filter(VClass == "ae"),
    color = ptol_red
  )+
  scale_x_continuous(
    transform = c("log10", "reverse")
  ) +
  scale_y_continuous(
    transform = c("log10", "reverse")
  )+
  coord_fixed()
```



Vowel Space

Remember how the resonant frequency of a tube is proportional to its length?

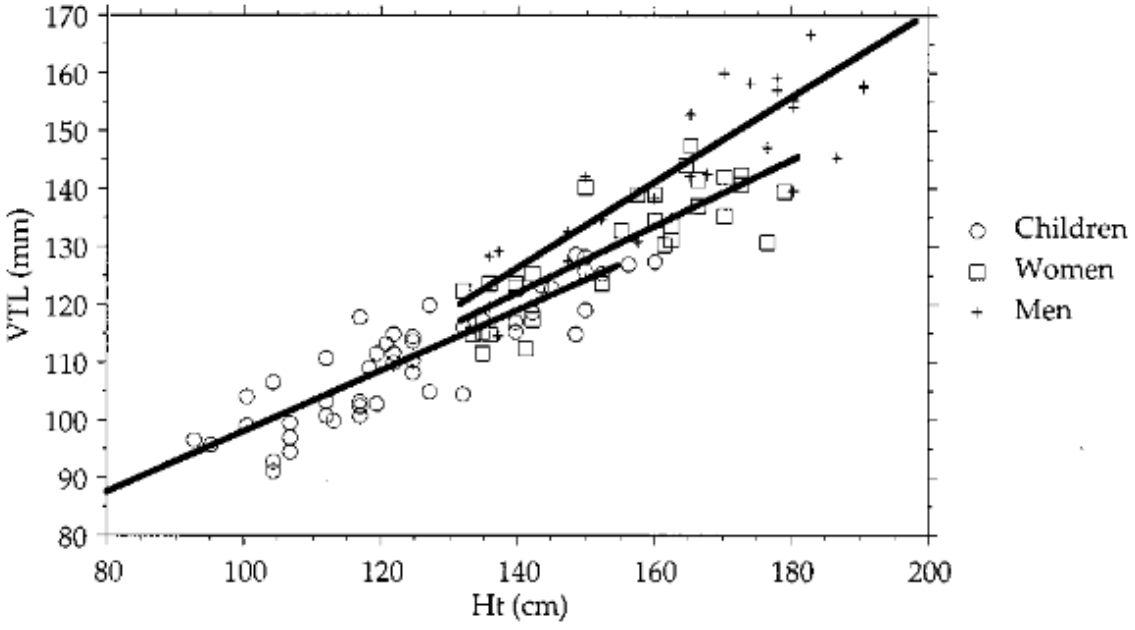
$$F_n = \frac{(2n-1)c}{4L}$$

```
expand_grid(
  formant = 1:3,
  vtl = c(14, 15, 16, 17)
) |>
  mutate(
    freq = (((2*formant)-1)*34300)/(4*vtl)
  ) |>
  mutate(
    vtl = str_glue("{vtl} cm"),
    formant = str_glue("F{formant}")
```

```
) |>
pivot_wider(
   names_from = vtl,
   values_from = freq
) |>
tt() |>
format_tt(
  i = 1:3,
  digits = 1,
) |>
style_tt(
  i = 1:3,
  align = "r"
)
```

formant	14 cm	15 cm	16 cm	17 cm
F1	612	572	536	504
F2	1838	1715	1608	1513
F3	3062	2858	2680	2522

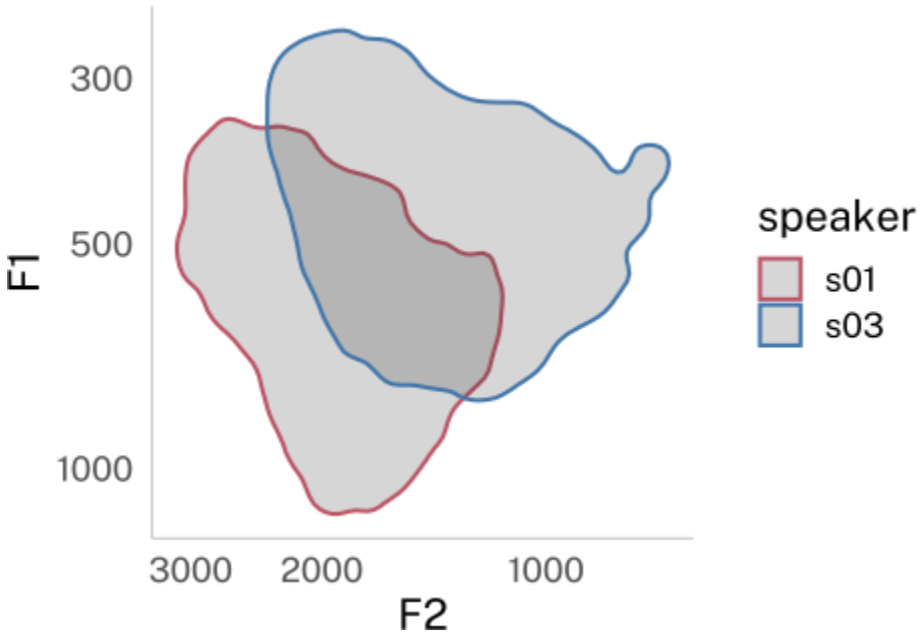
Vowel Space



Fitch & Giedd (1999)

Vowel Space

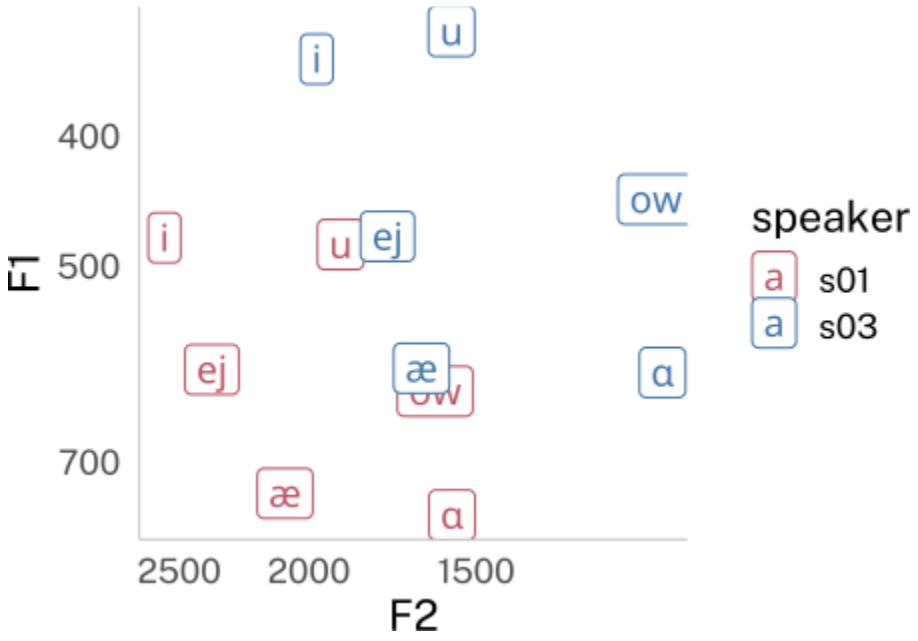
```
speaker_data |>
  ggplot(
    aes(F2, F1)
  )+
  stat_hdr(
    probs = 0.95,
    aes(
      color = speaker
    ),
    alpha = 0.2
  ) +
  scale_x_continuous(
    transform = c("log10", "reverse")
  ) +
  scale_y_continuous(
    transform = c("log10", "reverse")
  )+
  coord_fixed()
```



Vowel Space

```
speaker_data |>
  summarise(
    .by = c(ipa_vclass, plt_vclass, speaker),
```

```
  across(F1:F2, mean)
) |>
filter(
  plt_vclass %in% c(
    "iy", "ey", "ae",
    "o", "ow", "uw"
  )
) |>
ggplot(
  aes(F2, F1)
)+
geom_label(
  aes(
    label = ipa_vclass,
    color = speaker
  ),
  size = 5,
  family = "Noto Sans"
)+
scale_x_continuous(
  transform = c("log10", "reverse")
) +
scale_y_continuous(
  transform = c("log10", "reverse")
)+
coord_fixed()
```



Vowel Space

From the Philadelphia Neighborhood Corpus

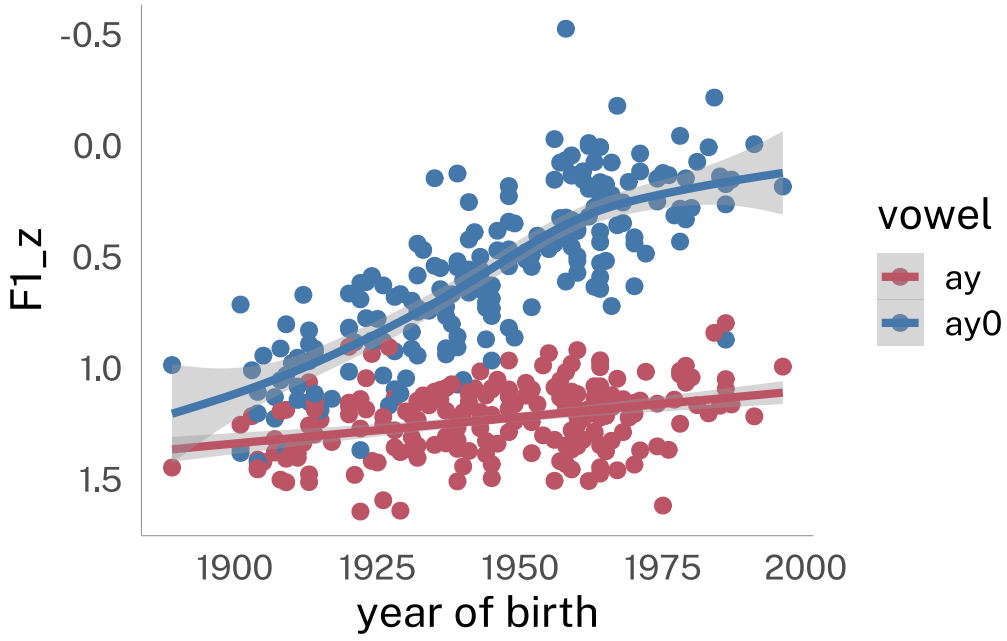
```
pnc_points <- read_csv("~/OneDrive - University of Kentucky/pnc_tidy/data/points/pnc_points.csv")
```

```
Rows: 751050 Columns: 31
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (16): idstring, ethnicity, sex, years_of_schooling, vowel, plt_vclass, i...
dbl (14): age, year, dob, stress, F1, F2, F3, B1, B2, B3, t, beg, end, dur
lgl  (1): not_phl

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
pnc_points |>
  filter(
    !not_phl,
    str_detect(ethnicity, "[ah]", negate = TRUE)
  ) |>
  norm_lobanov(
    .by = idstring,
    F1:F3
  ) |>
  filter(
    plt_vclass %in% c("ay", "ay0")
  ) |>
  summarise(
    .by = c(idstring, dob, plt_vclass, ipa_class),
    across(ends_with("_z"), mean)
  ) |>
  ggplot(
    aes(
      dob, F1_z, color = ipa_class
    )
  )+
  geom_point()+
  stat_smooth(method = gam, formula = y ~ s(x))+
  labs(
    color = "vowel",
    x = "year of birth"
```

```
  )+
  scale_y_reverse()
```

```
Normalization info
• normalized with `tidynorm::norm_lobanov()`
• normalized `F1`, `F2`, and `F3`
• normalized values in `F1_z`, `F2_z`, and `F3_z`
• grouped by `idstring`
• within formant: TRUE
• (.formant - mean(.formant, na.rm = T))/(sd(.formant, na.rm = T))
```



References

Bibliography

Fitch, W. T., & Giedd, J. (1999). Morphology and development of the human vocal tract: A study using magnetic resonance imaging. *The Journal of the Acoustical Society of America*, *106*(3), 1511–1522. https://doi.org/10.1121/1.427148